

SNOW GUARDIANS

SNOW GUARDIANS

Tech. Safety. Rescue.

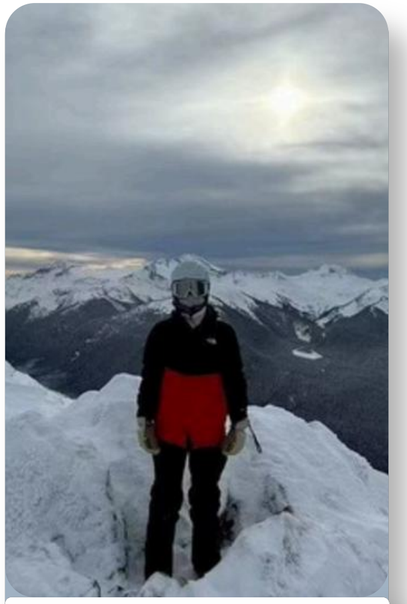
ZICHEN(KEVIN) XU | Kang Chiao International School
EMMA LIANG | International School Bangkok
RUIGUO(RAY) ZHAO | Shanghai World Foreign Language Academy
LINHAO(JOHN) LI | Qingdao Academy

Meet Our Team

SNOW GUARDIANS

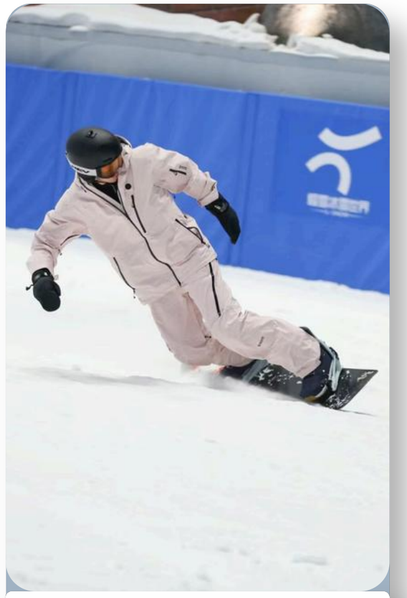
Empower the Ride,
Gaurdians By Your Side

We’re all skiers



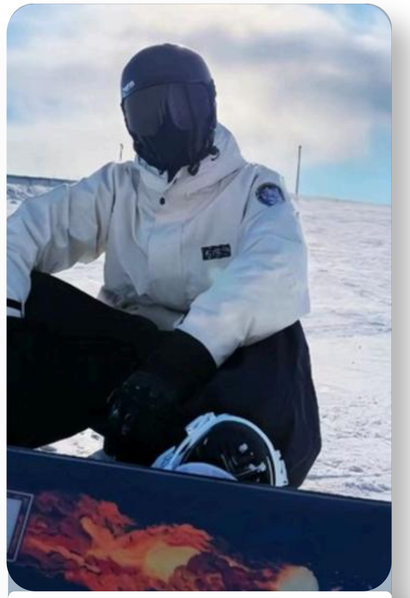
Emma Liang

Risk Evaluator &
Market Researcher



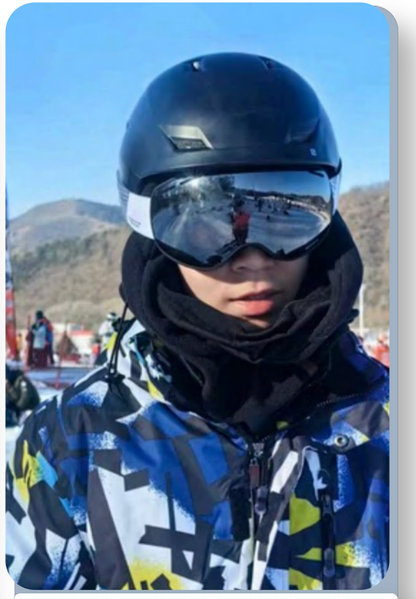
Kevin Xu

Product Lead &
Market Researcher



John Li

Product Marketing
Manager



Ray Zhao

Financial Analyst &
Sales Lead

THE PROBLEM

SNOW GUARDIANS



38 %

Head injuries comprise up to 38% of all skiing injuries and 29% of snowboarding injuries.



45 km/h

Average speeds reach 43.8 - 45.8 km/h (skiers) and 38.9-40.0 km/h (snowboarders).



4-5 mins

Whistler SAR records 90 “unable to locate” calls. The Golden Rescue Time is only 4-5 minutes.

Value Proposition & Differentiation



PILLAR PANEL

01. SAFETY

Collision detection, fall sensing, precise GPS, offline SOS.

PILLAR PANEL

02. INTELLIGENCE

IMU posture analysis, AI coaching, real-time feedback, web data dashboard.

PILLAR PANEL

03. ADAPTABILITY

Cold resistance, auto defog, solar hybrid power.

OUR CUSTOMER

SNOW GUARDIANS

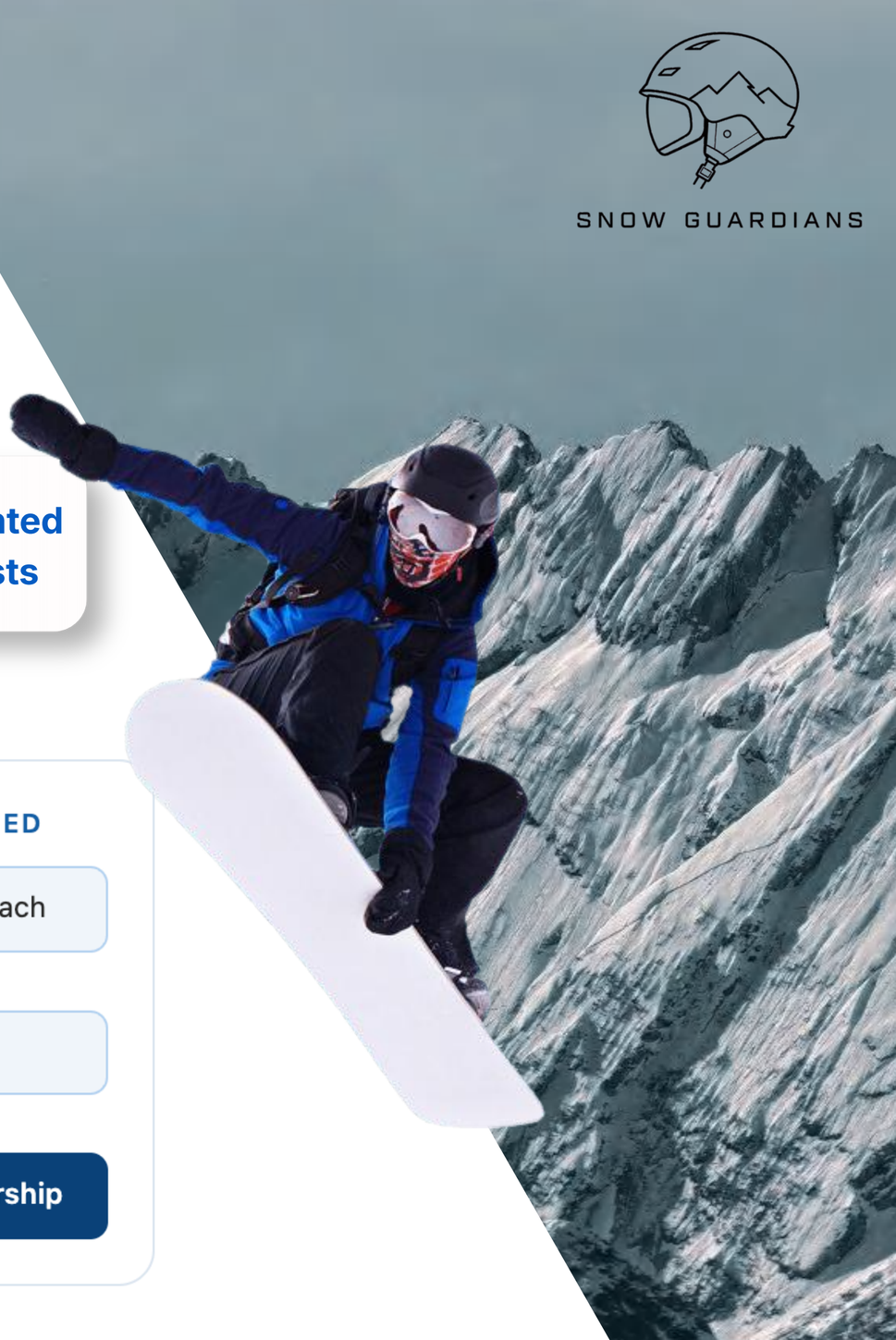
16+
Age Group

$7000
Income

All Skiers (Professional/ Beginner)

Injury-Anxious User

Tech-Oriented Enthusiasts



PAIN POINTS

- Fear of re-injury
- Lack of real-time safety awareness and control

TECH & SAFETY FOCUSED	PERFORMANCE FOCUSED
Recommendation	Professional network / coach
Safety research (device & app)	Product trial / demo
In-store purchase	**Direct purchase / sponsorship**

Market Opportunity

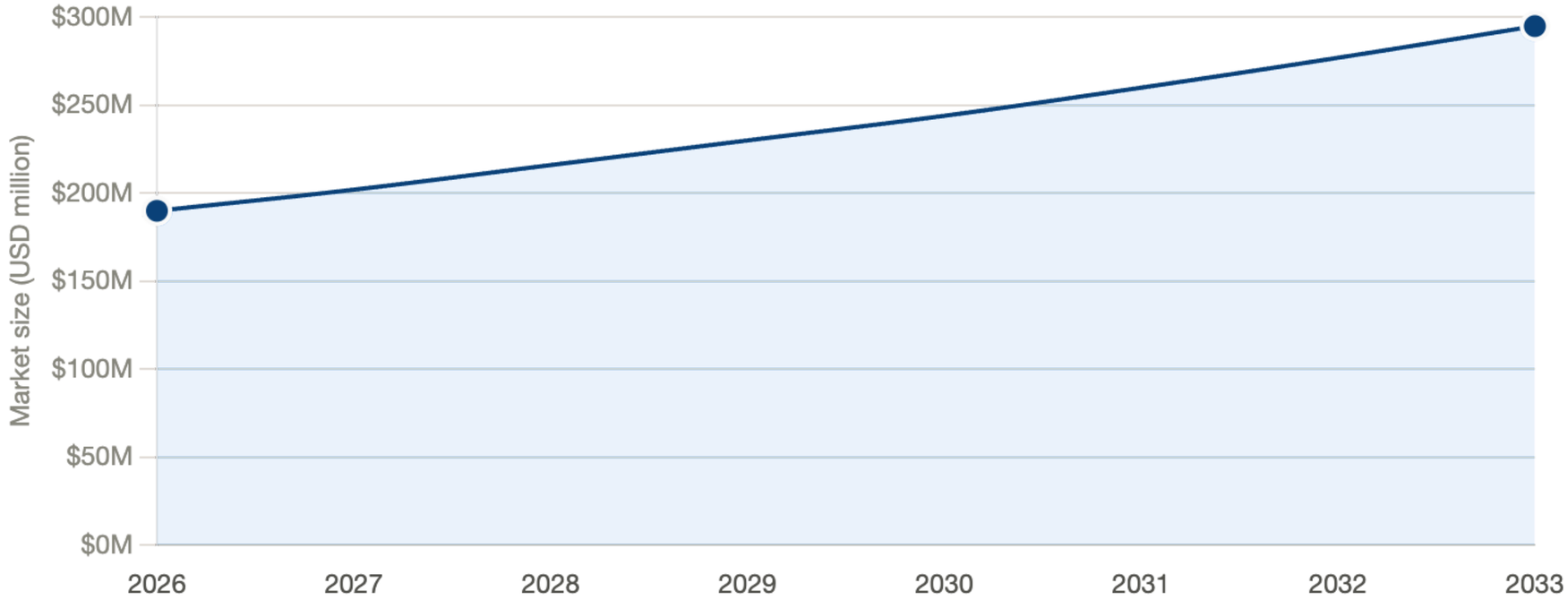
SNOW GUARDIANS

CORE DEMAND Drivers

- Younger riders drive demand (65% aged 15–35)
- Smart gear adoption inflecting (2.3% → 8.7% since 2021)
- Safety going from reactive to preventive

$385 Million
Market Size(2024)

6.5%
Forecasted CAGR(2026-2033)



77 Million
Global Helmet Users

8.7%
Smart gear penetration (2025)

User Journey

01

PHASE 1: BEFORE (The Summit)

Pair phone, set safety thresholds, receive AI pre-ski encouragement.

02

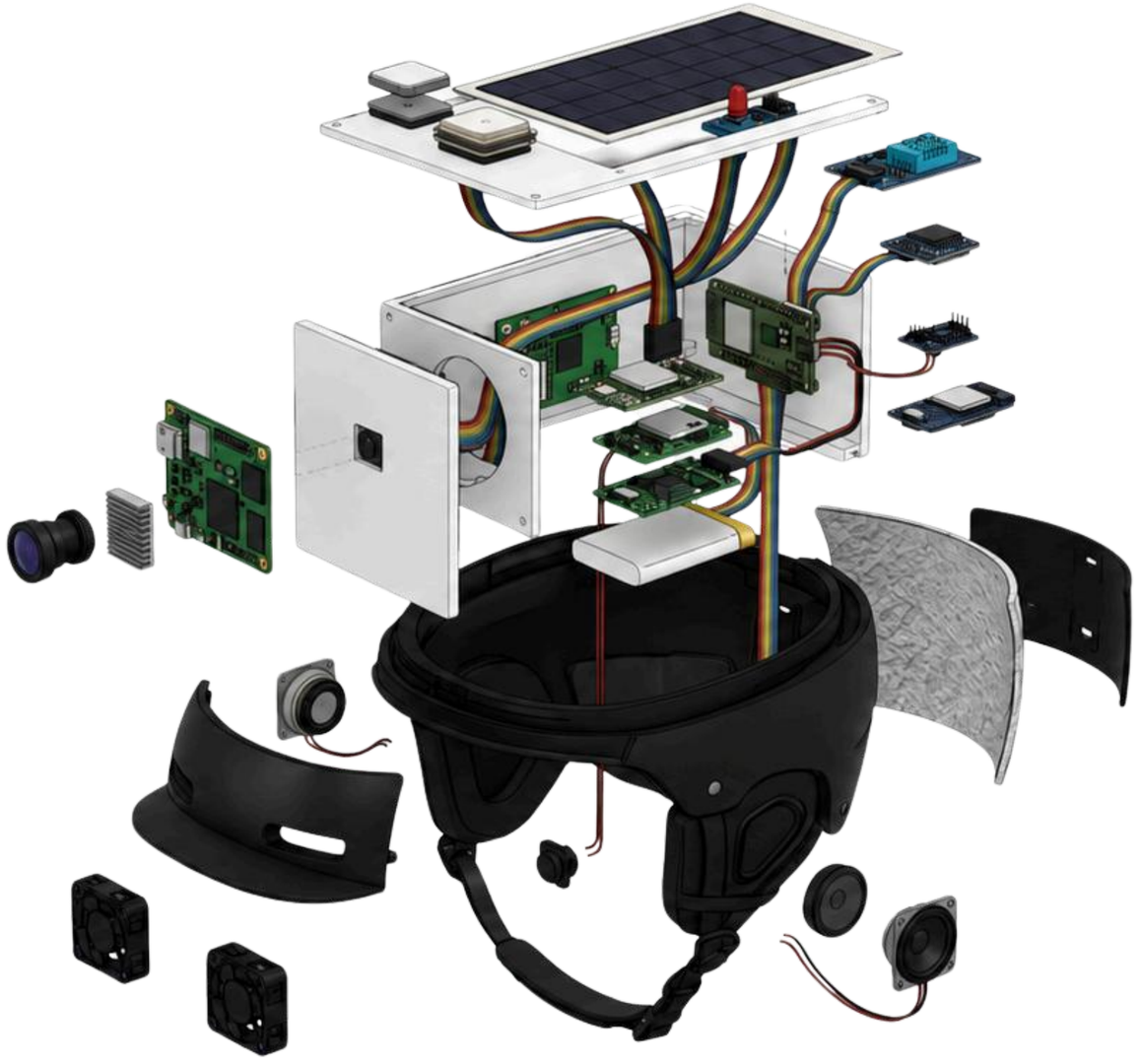
PHASE 2: DURING (The Descent)

Solar & battery charging active.
Obstruction & high-speed alerts.
GPS tracking, auto-defog, immediate SOS calls, AI guidance.

03

PHASE 3: AFTER (The Lodge)

AI-generated improvement suggestions based on recorded data.
Post-ski encouragement.



Technology Stack & Technical Specifications

SNOW GUARDIANS

SYSTEM ARCHITECTURE

Layer 1

Sense & Feedback

(The Nerves)

Layer 2

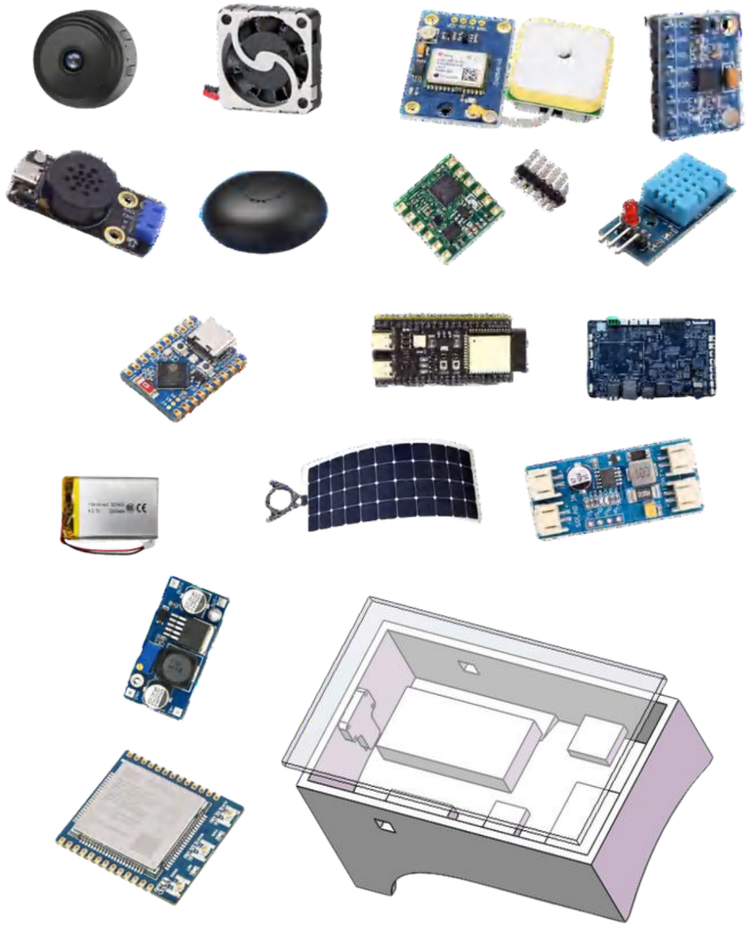
Core Compute & Power

(The Brain)

Layer 3

Cloud & Audio

(The Communication)



TECHNICAL SPECIFICATIONS

Core Computing & Control

RK3566 edge module, ESP32-S3 N16R8 microcontroller, YOLOv11 object detection model, Arduino IDE + VS Code embedded development

Sensing & Feedback Response

GPS module (speed & location), IMU (accelerometer + gyroscope for posture/angle), Temperature & humidity sensor, Camera for obstacle detection, Bluetooth audio alerts, Micro-fans for auto-defog

Communication & User Interaction

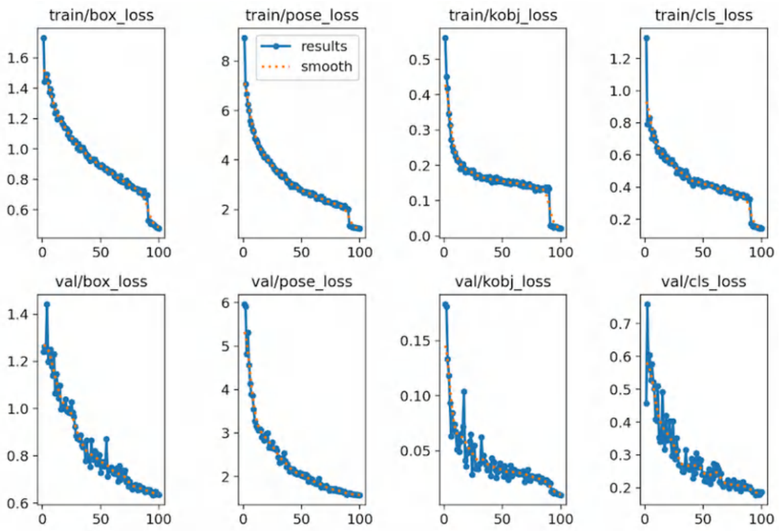
4G data transmission, Wi-Fi for web dashboard, Bluetooth audio, Bemfa cloud server, SOS alert to WeChat, LLM APIs, Web interface (session data, history, AI suggestions)

Power & Mechanical Design

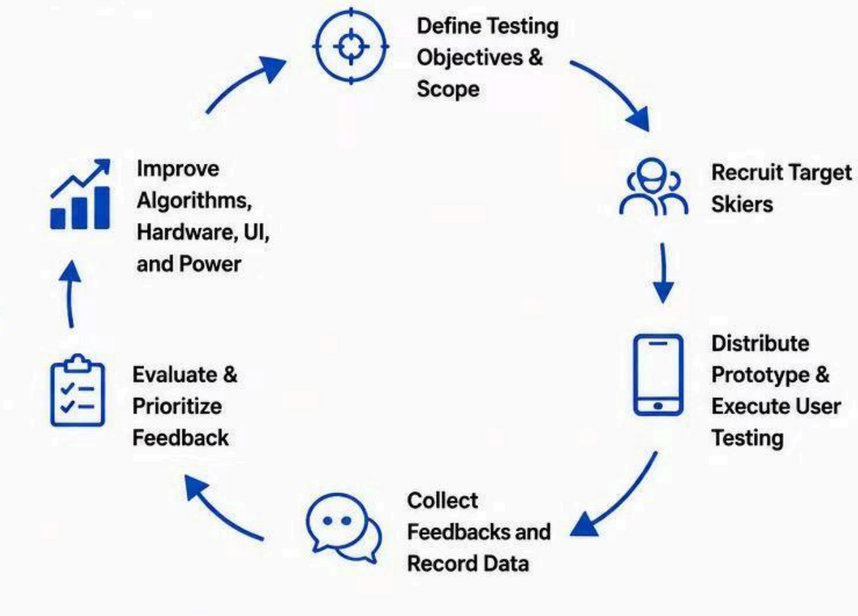
Lithium battery + flexible solar panel, Hybrid solar-battery power system, Boost converter, Thermal insulation layer, SolidWorks 3D modeling, 3D-printed PLA housing

Minimum Viable Product & Iteration

SNOW GUARDIANS







Available on MVP

- GPS Speed & Location
- AI IMU Analysis
- Immediate SOS Alert
- Obstacle Detection

Available on all models

- Auto Anti-Fog System
- Web & App UI

ALTITUDE: 2,880

SPEED: 55 KM/H

THE AI PARADIGM SHIFT

Core Engine:
Integrates three core functions: safety warnings, immediate SOS, and AI interaction/suggestions.

User Pain Points:
Dificulties in rescue, fear of injuries, and need for data-based skiing/snowboarding suggestions.

Algorithmic Power:
Unparallel algorithmic power: Two microcontrollers and one edge computing module supported by AI training model, optimizing accuracy and operational efficiency.



BLUE OCEAN STRATEGY

Eliminate:
Single-function helmets, short-range sensors, and extra gadgets.

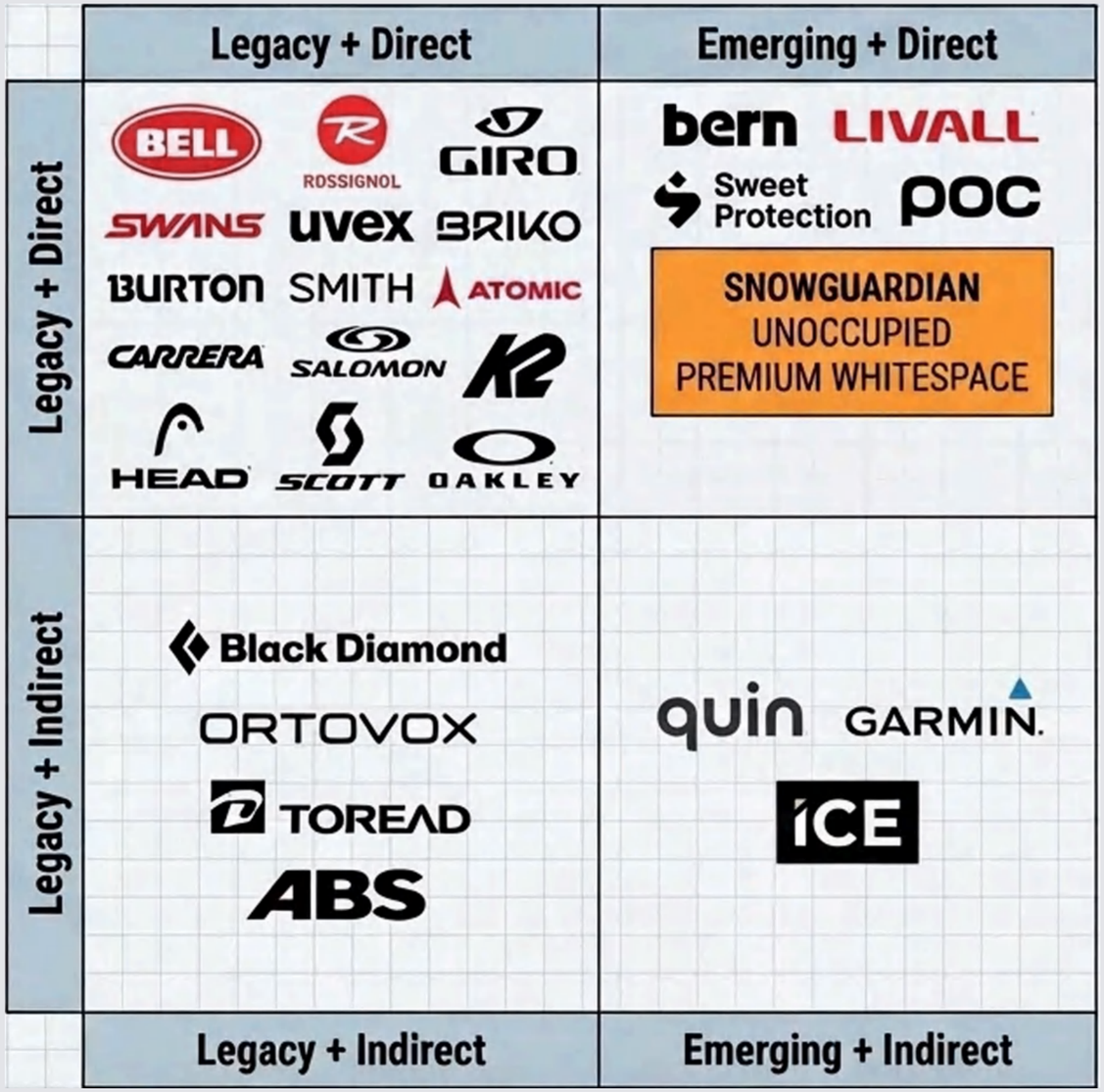
Raise:
Safety, visibility, responsiveness, and environmental reliability.

Reduce:
Manual checks, cold/moisture risks, and need for frequent charging.

Create:
Integrated helmet ecosystem with long-range detection, intuitive UI, hybrid power, and a new wearable-sports market.

Competitive Matrix & Specification Breakdown

SNOW GUARDIANS

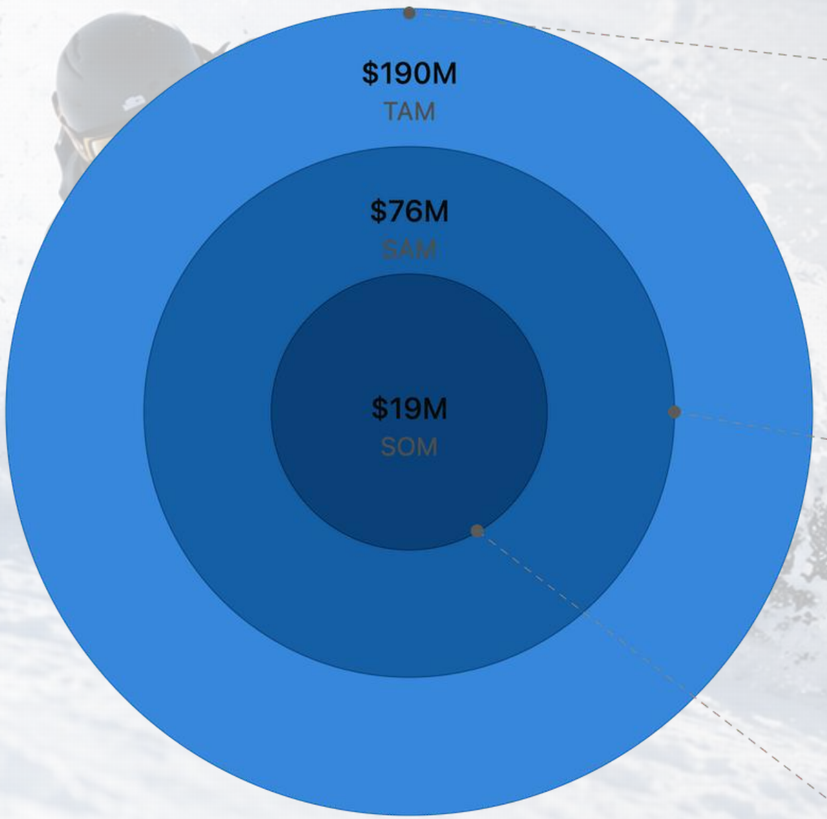


COMPETITOR SPECIFICATION BREAKDOWN

Dimension	SnowGuardian $300	Garmin $200–1000+	POC $150–300	LIVALL $220
Core Positioning	AI-powered soft-hard fusion for ski safety	GPS / Wearable device	High-end ski protective gear	Smart cycling helmet
Average Price	~$300	$200–1000+	$150–300	~$220
Ski-Specific Helmet	✅	❌	✅	❌
Offline Precision GPS + SOS	✅ Independent offline	⚠ inReach independent + subscription	❌	⚠ Phone-dependent
AI Crash Detection ≥92%	✅	❌	❌	⚠ Basic fall detection
IMU Posture / AI Coaching	✅	⚠ Wrist-based, not ski-optimized	❌	❌
Gross Margin	~54%→70%	58.7%	~45–50%	~30–38%
Net Margin	~29%	~22.6%	~8–12%	~3–8%
Market Size	Initial	$6.3B	~$35M→~$60–80M	~$30–50M
Annual Sales Volume	Year 1 ~63,000 units	18 million+ all categories	~800,000–1.2 million units	~150,000–300,000 units

Competitive Comparison & Marketing

SNOW GUARDIANS



TOTAL ADDRESSABLE MARKET (TAM)

$190 million

SERVICEABLE AVAILABLE MARKET (SAM)

$76 million

SERVICEABLE OBTAINABLE MARKET (SOM)

$19 million

25% of SAM — target achieved by 2029

TRADITIONAL VS. SMART

Metric	Traditional Helmet	Smart Helmet	Quantitative Advantage
Safety (Rescue)	No positioning, ~30% manual rescue success	≤5m GPS positioning, 70–75% rescue success	+40–45pp rate, 30% faster
Safety (Alerts)	No detection system	≥92% collision/fall detection, <5% false alarms	92% high-risk detected automatically
Intelligence (Coaching)	No performance insights	AI posture analysis + real-time feedback	Data-driven skill improvement
Environmental (Battery)	No electronics	≥8h cold-resistant battery +20% solar boost	Full-day smart functionality
Comfort	6.2/10 user rating	8.5/10 user rating	~37% comfort improvement

MARKETING STRATEGY

Brand Positioning: Premium smart safety helmet for high-end & professional skiers, blending safety, performance and tech-lifestyle design.

Channels:

Offline: Premium ski resort trial zones + ski club/team bulk partnerships.

Online: Ski KOL endorsements, short-video social media (WeChat/RedNote/TikTok/IG/YouTube).

Sales: Official website + high-end sports retail distribution.

Go-to-Market

SNOW GUARDIANS

PHASED ROADMAP

- **Phase 1 · Q2 2026:** Conceptual design & feasibility, define target customer, alpha prototype
- **Phase 2 · Q3-Q4 2026:** Prototype→MVP technical validation, structured beta, GB certification ▸ **Seed Round**
- **Phase 3 · Q1 2027:** Real-world user beta & compliance, customer feedback, brand e-commerce ▸ **Pre-A Round**
- **Phase 4 · Q2-3 2027:** Market-entry validation, soft launch (a few thousand units)
- **Phase 5 · Q4 2027:** large-scale production, national ad campaign, Asia ski resorts & retail ▸ **Series A**
- **Phase 6 · Q1 2028:** Cost & production optimization, material recycling, market feedback driving new features
- **Phase 7 · Q2 2028:** ▸ CPSC/CE international certs, Foreign ski-resort partnerships, enter North America/Europe ▸ **Series B**

FUNDING

Seed-round: $2M

Pre-A: $3M

Series A: $10M

Series B: $12M

Go-to-Market

SNOW GUARDIANS

PHASED ROADMAP

- **Phase 1 · Q2 2026:** Conceptual design & feasibility, define target customer, alpha prototype
- **Phase 2 · Q3-Q4 2026:** Prototype→MVP technical validation, structured beta, GB certification ▸ **Seed Round**
- **Phase 3 · Q1 2027:** Real-world user beta & compliance, customer feedback, brand e-commerce ▸ **Pre-A Round**
- **Phase 4 · Q2-3 2027:** Market-entry validation, soft launch
- **Phase 5 · Q4 2027:** large-scale production, national ad campaign, Asia ski resorts & retail ▸ **Series A**
- **Phase 6 · Q1 2028:** Cost & production optimization, material recycling, market feedback driving new features
- **Phase 7 · Q2 2028:** ▸ CPSC/CE international certs, Foreign ski-resort partnerships, enter North America/Europe ▸ **Series B**

FUNDING

Seed-round: $2M

Pre-A: $3M

Series A: $10M

Series B: $12M

Go-to-Market

SNOW GUARDIANS

PHASED ROADMAP

- **Phase 1 · Q2 2026:** Conceptual design & feasibility, define target customer, alpha prototype
- **Phase 2 · Q3-Q4 2026:** Prototype→MVP technical validation, structured beta, GB certification ▸ **Seed Round**
- **Phase 3 · Q1 2027:** Real-world user beta & compliance, customer feedback, brand e-commerce ▸ **Pre-A Round**
- **Phase 4 · Q2-3 2027:** Market-entry validation, soft launch
- **Phase 5 · Q4 2027:** large-scale production, national ad campaign, Asia ski resorts & retail ▸ **Series A**
- **Phase 6 · Q1 2028:** Cost & production optimization, material recycling, market feedback driving new features
- **Phase 7 · Q2 2028:** ▸ CPSC/CE international certs, Foreign ski-resort partnerships, enter North America/Europe ▸ **Series B**

FUNDING

Seed-round: $2M

Pre-A: $3M

Series A: $10M

Series B: $12M

Go-to-Market

SNOW GUARDIANS

PHASED ROADMAP

- **Phase 1 · Q2 2026:** Conceptual design & feasibility, define target customer, alpha prototype
- **Phase 2 · Q3-Q4 2026:** Prototype→MVP technical validation, structured beta, GB certification ▸ **Seed Round**
- **Phase 3 · Q1 2027:** Real-world user beta & compliance, customer feedback, brand e-commerce ▸ **Pre-A Round**
- **Phase 4 · Q2-3 2027:** Market-entry validation, soft launch
- **Phase 5 · Q4 2027:** large-scale production, national ad campaign, Asia ski resorts & retail ▸ **Series A**
- **Phase 6 · Q1 2028:** Cost & production optimization, material recycling, market feedback driving new features
- **Phase 7 · Q2 2028:** ▸ CPSC/CE international certs, Foreign ski-resort partnerships, enter North America/Europe ▸ **Series B**

FUNDING

Seed-round: $2M

Pre-A: $3M

Series A: $10M

Series B: $12M

Activity Map

SNOW GUARDIANS

SNOW GUARDIANS

High-Performance Precision Sensing

- Choose premium sensors over lowest cost (accuracy over margin)
- Feed IMU posture data back into model iteration
- 92% accuracy as ship/no-ship gate
- Train on proprietary accident dataset(YOLOv11 Model)
- Assemble best-in-class components

Rapid Emergency Response Infrastructure

- Offline-first architecture — no dependence on phone/network
- Multi-channel alerting (4G + Bluetooth) for redundancy
- merge into Rapid Emergency Response (it's an output, not a decision)
- Conservative false-alarm threshold

Integrated Data & AI Ecosystem

- Tiered subscription monetization
- Retain user data to power the data flywheel

Adaptive EnvironmentEngineering

- Choose solar-hybrid power to solve the cold-weather battery trade-off
- merge under Adaptive Environment Engineering
- Small-batch production (500–1,000 units)

Cost Structure

HARDWARE (CORE)	
RK3566 + camera	$76.25
ESP32-S3	$4.16
Sensors (temp + MPU6050)	$3.33
GPS module	$6.81
4G module	$3.44
Bluetooth earphone	$5.83
POWER SYSTEM	
Battery fans	$4.15
Battery (2000mAh)	$1.56
Solar panel	$3.06
Solar charging module	$0.75
Voltage regulator	$2.50
MECHANICAL	
Helmet	$19.89
Metal cooling case	$5.00
Insulation cotton	$0.81
AUXILIARY & MISC	
Cooling fans (x2)	$3.06
Voice module (SU03)	$4.89
Wires	$0.39
Unit cost of prototype	**$147.67**

SNOW GUARDIANS

VOLUME SCALING EFFICIENCY

PRODUCTION VOLUME	UNIT COST	REASON
100 units	High	Low efficiency
500 units	Medium	Bulk purchasing
1000+ units	Low	**Optimized production**

COGS per unit

DRIVER / LINE ITEM	2027	2028	2029	2030
Units sold	8,000	50,000	110,000	220,000
Prototype BOM ($/unit)	$147.67	$147.67	$147.67	$147.67
Component cost factor	92%	80%	74%	69%
Freight & duties (% of component)	5%	5%	5%	5%
COST BREAKDOWN				
Component cost (BOM × factor)	$135.86	$118.14	$109.28	$101.89
Assembly & test labor	$12.00	$8.00	$6.00	$4.50
Packaging & accessories	$4.50	$3.50	$3.00	$2.50
Freight & duties	$6.79	$5.91	$5.46	$5.09
Warranty & returns reserve	$12.00	$7.00	$5.50	$4.50
Scrap / yield loss	$3.85	$2.45	$2.76	$1.52
COGS per unit	**$175.00**	**$145.00**	**$132.00**	**$120.00**

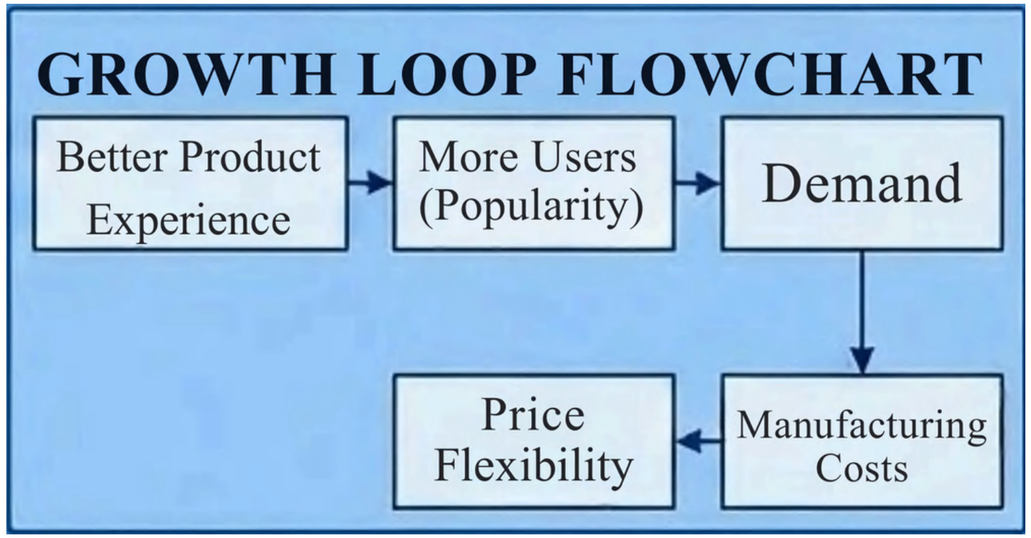
Business Model

SNOW GUARDIANS
TECH SAFETY. RESCUE.

Category	Item	2027	2028	2029	2030
Total Revenue		$2,486,256	$16,078,200	$38,383,400	$83,686,592
Revenue Growth Drivers	ASP	$300.00	$300.00	$310.00	$320.00
	Units Sold	8,000	50,000	110,000	220,000
	Sub Price/Month	$5.99	$5.99	$6.49	$6.99
	Sub Penetration	30%	45%	60%	72%
	Sub Duration	6	8	10	12
Operating Revenue	Hardware	$2,400,000	$15,000,000	$34,100,000	$70,400,000
	Subscription	$86,256	$1,078,200	$4,283,400	$13,286,592
	Total Operating Revenue	$2,486,256	$16,078,200	$38,383,400	$83,686,592
Costs of Goods Sold	COGS/Unit	$175.00	$145.00	$132.00	$120.00
	Total COGS	$1,400,000	$7,250,000	$14,520,000	$26,400,000
Gross Profit		$1,086,256	$8,828,200	$23,863,400	$57,286,592
Gross Margin		43.69%	54.91%	62.17%	68.45%
Operating Expenses	Total OpEx	$2,836,000	$8,068,000	$17,476,000	$31,020,000
Administration	Salaries	$120,000	$225,000	$364,000	$540,000
	Other Admin	$100,000	$170,000	$278,000	$405,000
	Staff Count	3	5	7	9
	Pay Per Count	$40,000	$45,000	$52,000	$60,000
	Business Trips	$20,000	$25,000	$30,000	$35,000
	Rent	$30,000	$30,000	$40,000	$50,000
	Total Admin	$270,000	$450,000	$712,000	$1,030,000
Research&Development	Total R&D	$1,284,000	$1,912,000	$2,564,000	$3,240,000
	Salaries	$384,000	$612,000	$864,000	$1,140,000
	Staff Count	12	18	24	30
	Pay Per Count	$32,000	$34,000	$36,000	$38,000
	Other Dev Costs	$900,000	$1,300,000	$1,700,000	$2,100,000
Sales&Marketing	Total S&M	$1,282,000	$5,706,000	$14,200,000	$26,750,000
	Salaries	$132,000	$506,000	$1,200,000	$2,250,000
	Staff Count	6	22	50	90
	Pay Per Count	$22,000	$23,000	$24,000	$25,000
	BD Costs	$350,000	$2,200,000	$5,500,000	$10,500,000
	BD Per Headcount	$58,333	$100,000	$110,000	$116,667
	Offline Marketing	$400,000	$1,500,000	$3,500,000	$6,500,000
	Online Marketing	$400,000	$1,500,000	$4,000,000	$7,500,000
Operating Income		($1,749,744)	$760,200	$6,387,400	$26,266,592
Break-Even	Fixed OpEx	$2,836,000	$8,068,000	$17,476,000	$31,020,000
	Contribution Margin/Unit	$125.00	$155.00	$178.00	$200.00
	Break-Even Units	22,688	52,052	98,180	155,100
Key Growth Metrics	Customer Acquisition Cost	$144	$104	$118	$111
	Customer Lifetime Value	$160	$207	$253	$297
	LTV:CAC Ratio	1.1x	2.0x	2.1x	2.7x
Tax Rate		15%	15%	15%	15%
Tax Expenses		-	$114,030	$958,110	$3,939,989
Net Income		($1,749,744)	$646,170	$5,429,290	$22,326,603

SUBSCRIPTION TIERS

BASIC	PREMIUM	PRO
$5.99/mo	**$11.99/mo**	**$17.99/mo**
Essential tracking Limited data storage Standard support	Advanced analytics Cloud backup Priority support	Full feature suite Unlimited storage Dedicated account manager



Social Return on Investment （SROI）

SNOW GUARDIANS

Social Outcomes #1

Reduce Deaths

Precision GPS + Offline SOS Network enables faster rescue (SAR) success rate from **~30% to 70-75%**

Social Outcomes #2

Reduce serious injuries

>92% Crash Detection & Fall Pre-warning

Social Outcomes #3

Reduce the burden on the rescue system

Pinpoint localization shortening rescue windows

Social Outcomes #4

Lowering psychological barriers & promoting healthy exercise

Eliminating the "fear of injury" barrier

Strategic SROI Maximization Levers

1.Bind rescue agencies and ski resorts to amplify the social leverage of each device

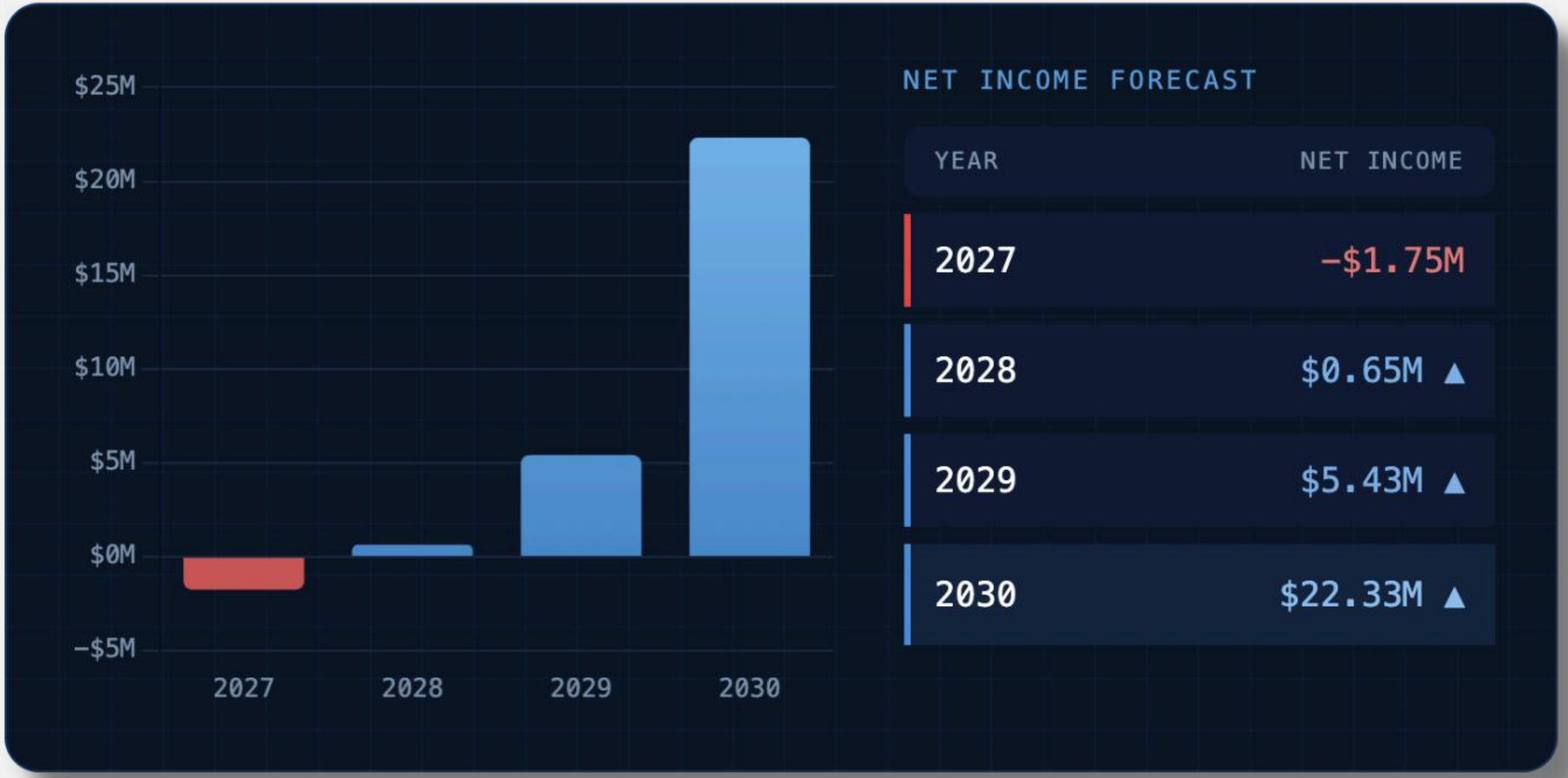
2.Improve penetration rate and subscription retention

3.Using economies of scale to reduce unit investment

4.Covering vulnerable/high-risk groups and enhancing additionality

Four Year Financial Projection

SNOW GUARDIANS



NET INCOME FORECAST

YEAR	NET INCOME
2027	−$1.75M
2028	$0.65M ▲
2029	$5.43M ▲
2030	$22.33M ▲

Risk Management

SNOW GUARDIANS

Technical Risk #1

Threats: AI misclassification and biased accident-risk prediction

Defence: Expand training dataset + continuous model retraining (YOLOv11); enforce ≥0.9 confidence threshold

Technical Risk #2

Threats: Core chipset (RK3566) shortage or single-supplier price shock

Plan A: Lock in bulk procurement agreement with primary supplier for cost advantage
Plan B: If disrupted, switch to pre-qualified backup supplier within 48h; temporarily scale down that quarter's output to protect paid orders first

Market Risk

Threats: Seasonal Demand Concentration (Winter-Dependent Sales)

Defence: Align sales & distribution with seasonal peaks across hemispheres
N. Hemisphere: Nov–Mar
S. Hemisphere: Jun–Sep

Legal Risk

Threats: lawsuits from users injured while using the helmet,

Defence: Require signed liability waiver at use and liability insurance to cover legal defense costs

SNOW GUARDIANS

Empower the Ride
Gaurdians By Your Side

- Collision detection + high-speed warning
- Environmental adaptability
- Precise GPS + Offline SOS
- AI algorithm
- Enhanced visibility

Raising Seed-round funding of $2M

40% R&D
40% SG&A
15% GTM
5% Contingency

Email: Snowguardians@gmail.com
Website: www.snowguardians.com